MUHAMMAD IHSAN RIJADIN

Surabaya, East Java | 085113505249 | ihsanrijadin07@gmail.com| **LinkedIn**: www.linkedin.com/in/muhammad-ihsan-680817400 | **Portfolio**: blue-sight01.github.io | **GitHub**: github.com/Blue-Sight01

PROFILE

Applied Bachelor (D4) Computer Engineering student at Politeknik Elektronika Negeri Surabaya (PENS) with a strong interest in Software Development, Database Management, and Hardware Engineering. Experienced in designing desktop applications with relational database integration, managing large-scale data, and leading technical teams. Highly adaptable and eager to contribute to the broader technology ecosystem, from software solutions to digital infrastructure.

TECHNICAL SKILLS

- **Software & Database:** C, C++, Python, PostgreSQL, Google Sheets Automation
- **Hardware & Embedded:** Microcontrollers (ESP32), Circuit Design & PCB Etching
- **Tools & Frameworks:** Microsoft Foundation Class (MFC), psqlODBC, AutoCAD, Xilinx ISE, Cisco Packet Tracer, Git/GitHub

PROJECT EXPERIENCE

Laundry Transaction Cashier Application | *Academic Project* | [GitHub Link]

- Developed a desktop cashier application using C++ and the Microsoft Foundation Class (MFC) framework.
- Integrated the application with a PostgreSQL relational database via psqlODBC (CDatabase & CRecordset) for persistent data management.
- Designed features for multi-item input per transaction, sequential navigation for history search, monthly filters, and financial report exports (.txt).

Coworking Space Database Management System | *Academic Project* | [GitHub Link]

- Designed an enterprise-scale relational database architecture (11 tables) using PostgreSQL for coworking space operations.
- Implemented business logic automation using Triggers and PL/pgSQL to prevent double-booking and update F&B stock in real-time.
- Built Role-Based Views to present specific data dashboards for Manager, Finance, and Marketing roles.

Hardware Prototyping (Voltmeter & Robo Soccer) | *PENS*

- Created robot chassis and component designs using AutoCAD.
- Executed hardware fabrication, including acrylic cutting with CNC 3018, circuit path printing (PCB etching), and cable wiring.

ORGANIZATIONAL & COMMITTEE EXPERIENCE

UKM RoboHolic PENS | *Project Manager & Technical Lead* | 2026

- Led a technical team in the design, assembly, and testing of hardware prototypes.
- Coordinated task delegation and performed on-site mechanical and electronic troubleshooting to ensure optimal prototype functionality.

LKMM TPD 2026 | *Facilitator & Data Center Committee* | Jul 2026 - Nov 2026

- Conceptualized and delivered training sessions for the Pre-Basic Student Management Leadership Training (LKMM TPD) PENS 2026.
- Developed an automated data processing system using advanced Google Sheets formulas to streamline reporting efficiency.
- Created and managed Google Groups for streamlined information distribution.

OSIS SMA Negeri 1 Pangkah | *Head of Class Meeting & Division 4 Coordinator* | 2023 - 2024

- Managed operations and logistics for school-wide events to ensure smooth execution.
- Ensured disciplined coordination across divisions according to established schedules and rundowns.

EDUCATION

Politeknik Elektronika Negeri Surabaya (PENS) | Surabaya, East Java

- Applied Bachelor (D4) in Computer Engineering | 2025 - Present

SMA Negeri 1 Pangkah | Tegal, Central Java

- Mathematics and Natural Sciences (MIPA) | Graduated 2025